



Periodic Patterns of Space Use in Animal Tracking Data

Continuous-Time Animal Movement Modeling

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Overview

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- 1) Introduction
- 2) Methods for visualizing periodic patterns on space use
- 3) Interpreting periodograms
- 4) Applications

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 - Cycles in resource availability
 - Memory effects and cognitive abilities
 - Social interactions

Periodicities

Periodic patterns of space use

Periodicity in activity levels

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Periodicities

Periodic patterns of space use

- Repeated visits to a given area with a characteristic schedule
 - “movement path recursion”
 - i.e., specific movement trajectories that occur at a generally predictable interval

Periodicity in activity levels

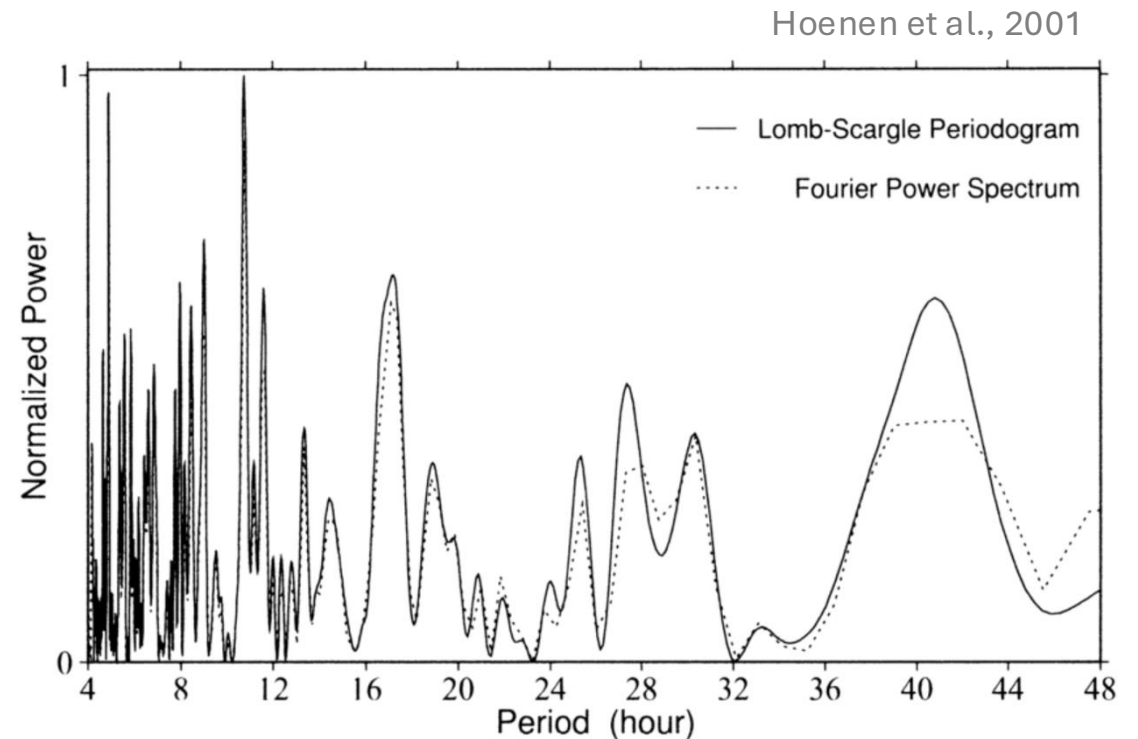
- Cycles in movement activity
 - Movement speed, turning angle, acceleration, physiological or behavioral state
 - i.e., rest/activity cycles

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Visualizing periodicities

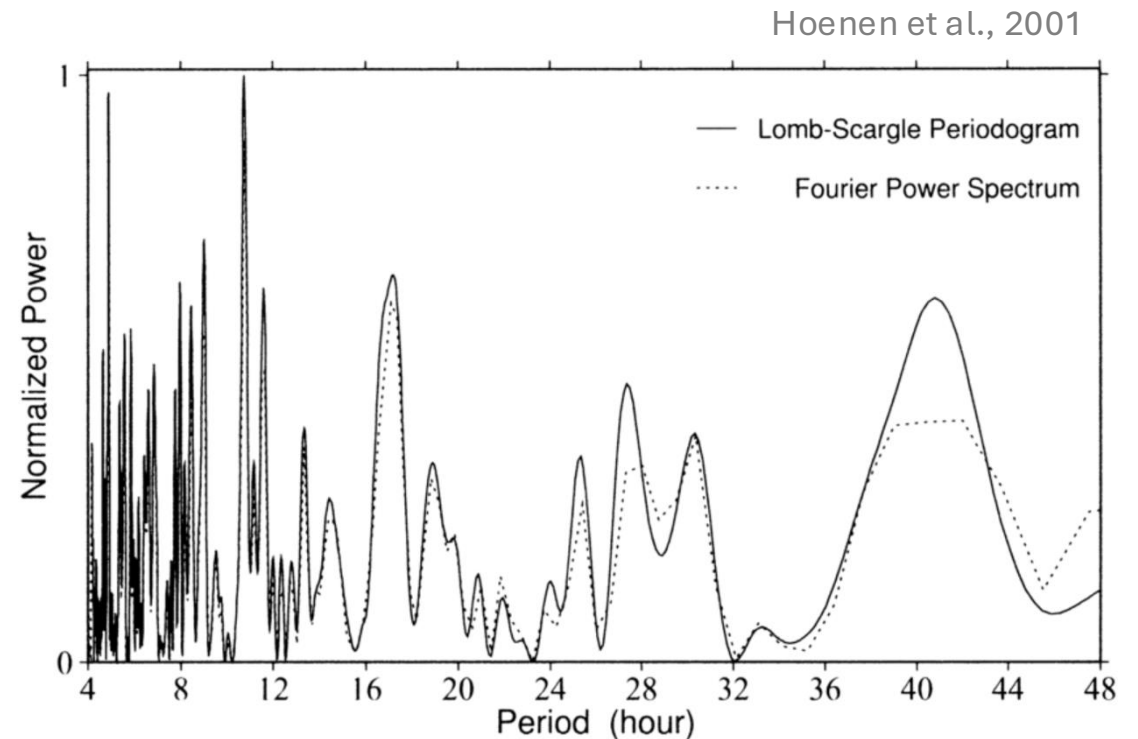
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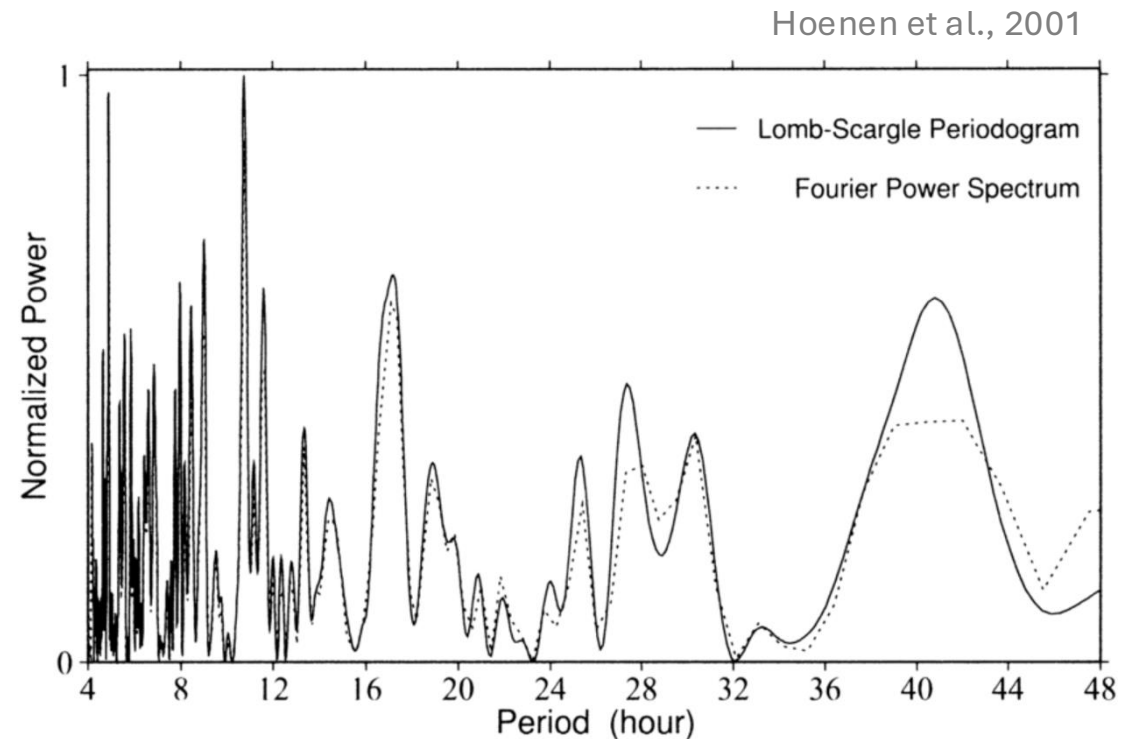
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 - Power as a function of frequency (or period, $1/\text{frequency}$)



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Periodograms decompose a signal (e.g., animal location data), and can help identify the most prominent frequencies in the signal

- Nonparametric test of periodicity
 - Power as a function of frequency (or period, $1/\text{frequency}$)
- Common in other disciplines



Visualizing periodicities

The **discrete Fourier transform** (DFT) yields an estimate of the contribution (strength) of any given frequency (f) to the signal:

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- Computed via fast Fourier transform (FFT)

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The **Lomb-Scargle periodogram** (LSP) is an extension of the DFT-periodogram that accounts for **irregular sampling or missing data**:

$$\widehat{LSP}^{(k)}(f) = \frac{DFT\{w\}(0) \cdot |DFT\{wX\}(f)|_k|^2 - \text{Re}(DFT\{w\}(2f) \cdot DFT\{wX\}(f)|_k|^2)}{FT\{w\}(0)^2 - |DFT\{w\}(2f)|_k|^2} \quad \widehat{LSP}(f) = \frac{1}{K} \sum_{k=1}^K \widehat{LSP}^{(k)}(f)$$

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- Missing data → uses intended sampling schedule
- “Duty cycled” data → uses finest sampling interval
- Variation around intended interval → defaults to median interval

Visualizing periodicities

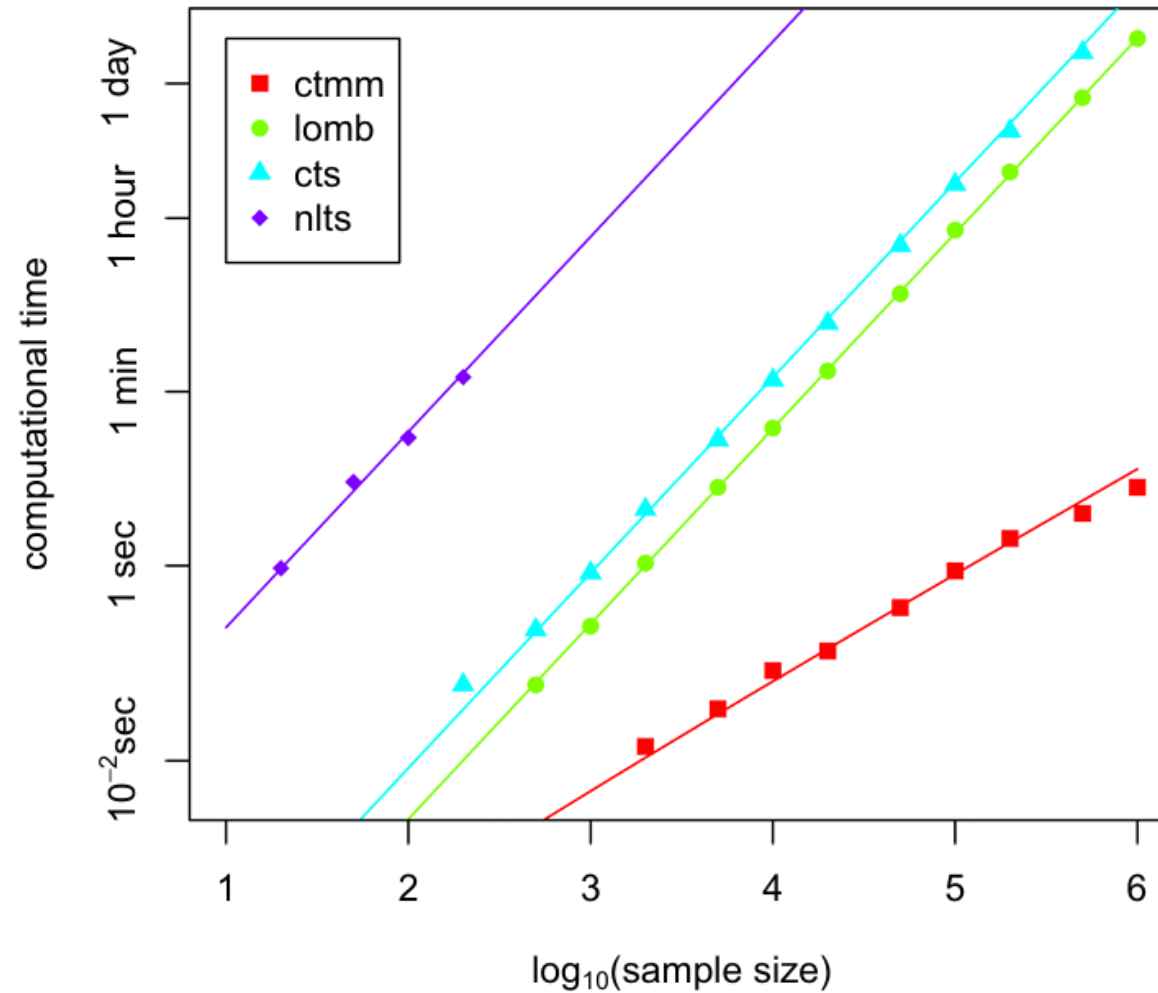
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- Computed via FFT $\rightarrow O(N \log N)$
 - *Based entirely on the DFT*
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 - Examine periodicity in longitude, latitude, altitude jointly or separately
- Multiple individuals
 - Fitting multi-individual periodograms can help augment sample size
 - Can better distinguish periodicity from background noise

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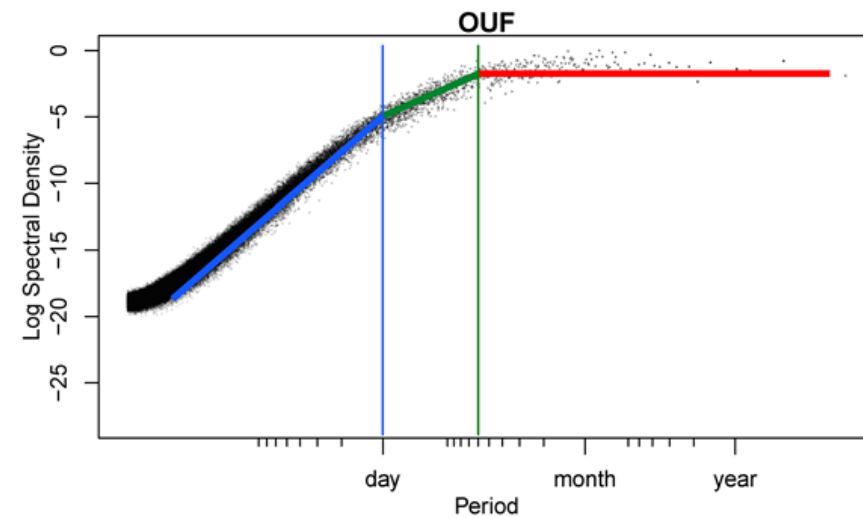
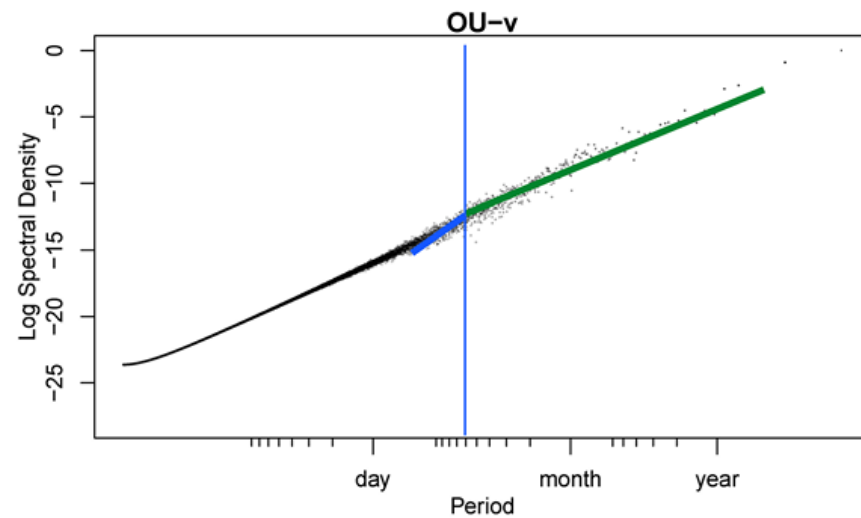
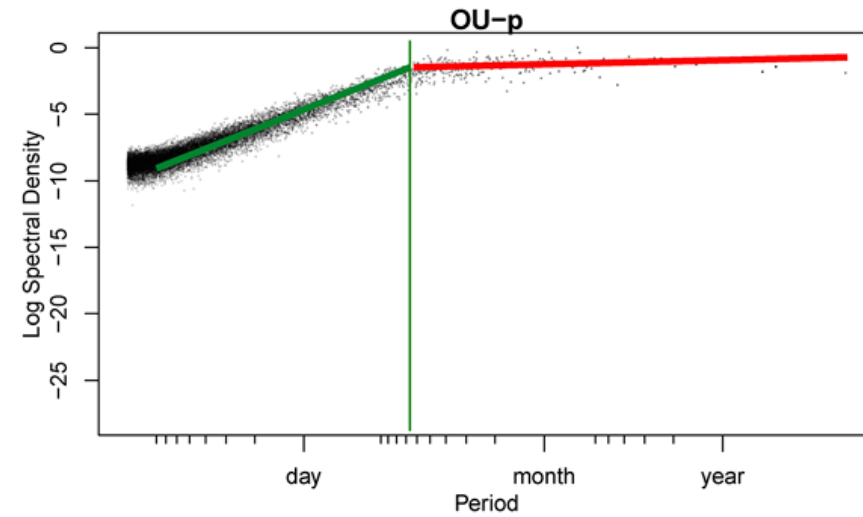
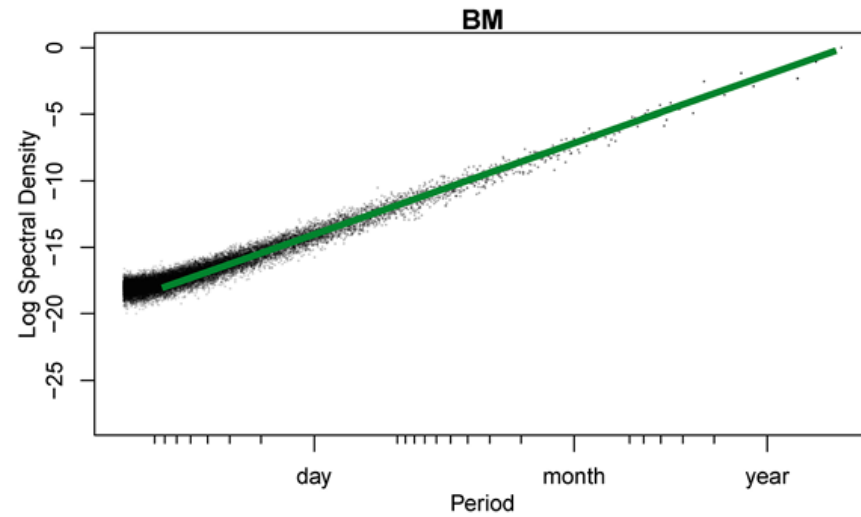
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- **Harmonic**: component frequency that is an integer multiple of the fundamental frequency (largest signal-to-noise ratio)
 - Occurs if waveform is not perfectly sinusoidal
 - Be careful of peaks at fractional periods of the fundamental period

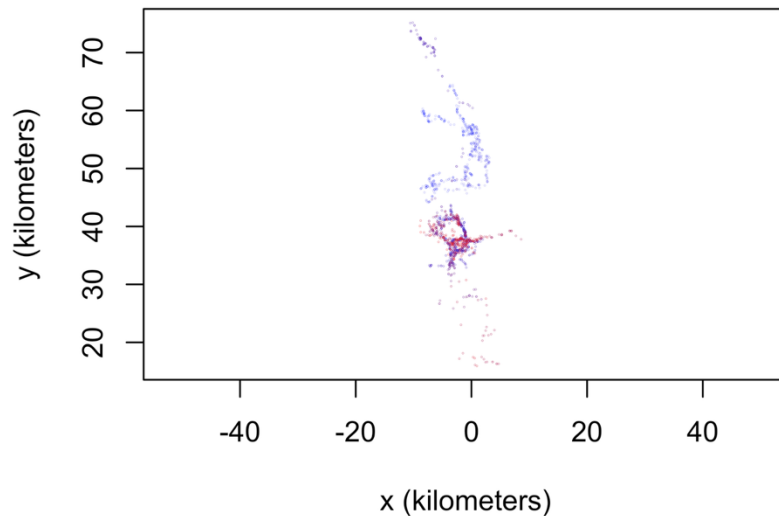
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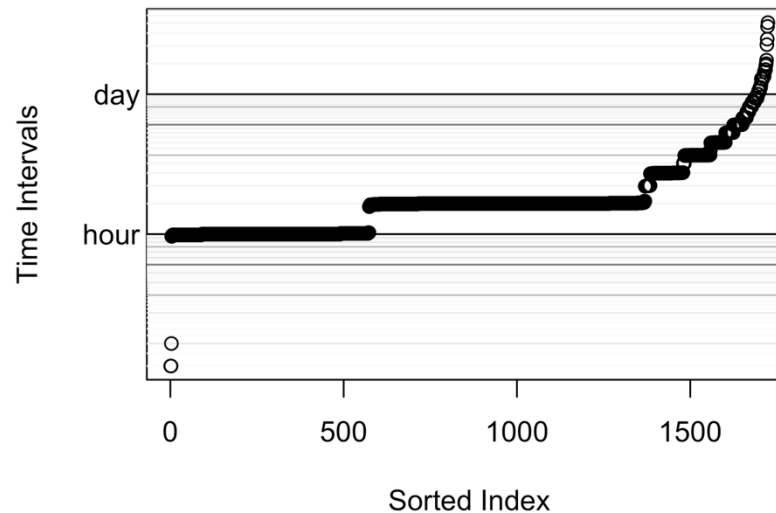
Interpreting periodograms

- Periodicities may occur as an **artifact of sampling schedule**
 - Duty cycling between fine/coarse sampling rates creates a periodic effect
 - Missing data may occur in an autocorrelated manner

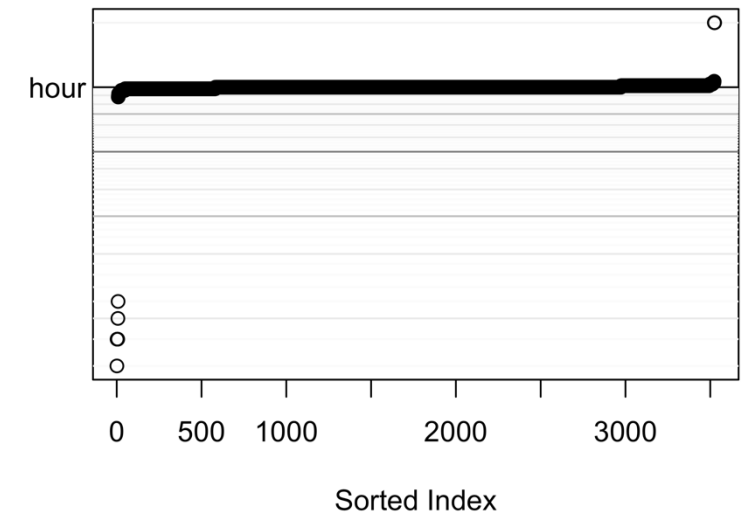
Individual with GPS Collar Malfunction



Sampling Intervals with Missing Data



Consistent Sampling Intervals



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 - 1) **Visual diagnostic:** Plot periodogram of location time series alongside periodogram of sampling schedule (potential false positive if same peak occurs in both)

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- **Identify false positives:**
 - 1) **Visual diagnostic:** Plot periodogram of location time series alongside periodogram of sampling schedule (potential false positive if same peak occurs in both)
 - 2) **Null model approach:** Aperiodic continuous-time stochastic model acts as null model to generate simulated datasets of the same sampling schedule as the real data (identify irregularities; compute proportion of simulations in which value at period of interest $>$ real data)

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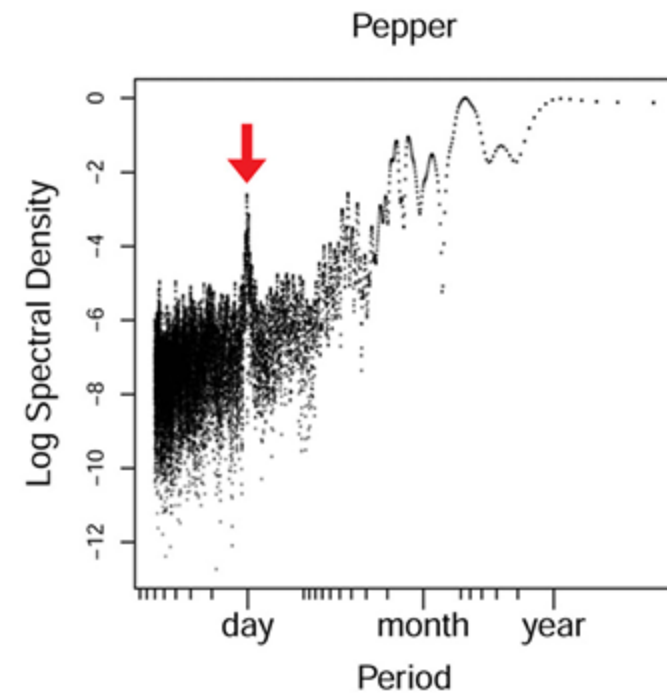
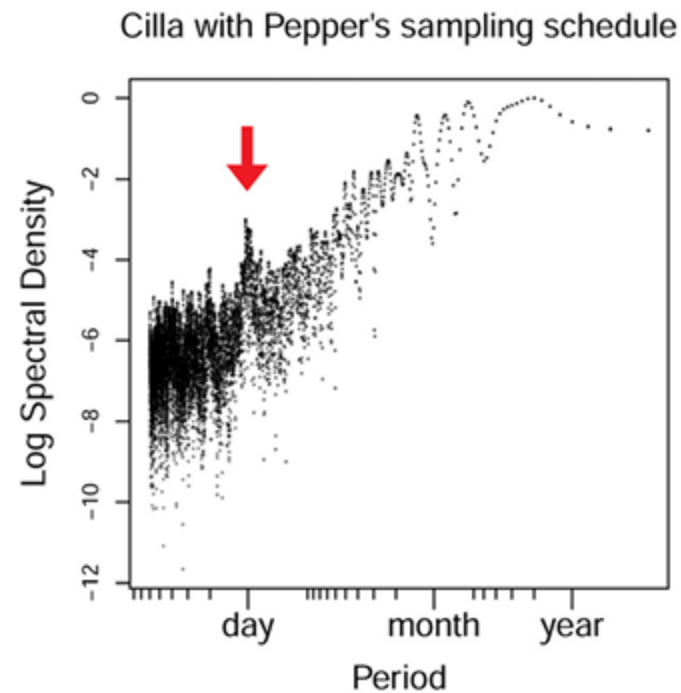
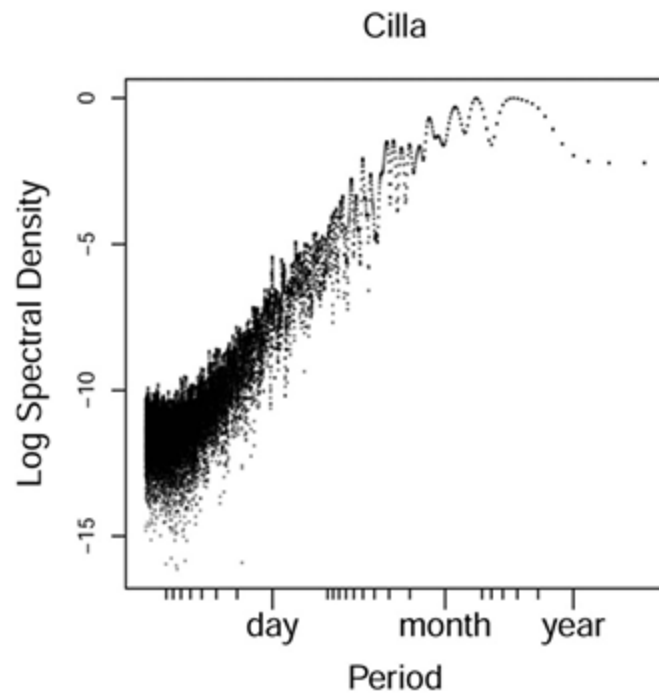
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Case studies

- African buffaloes: Cilla & Pepper
 - Sampling interval: 1 hour

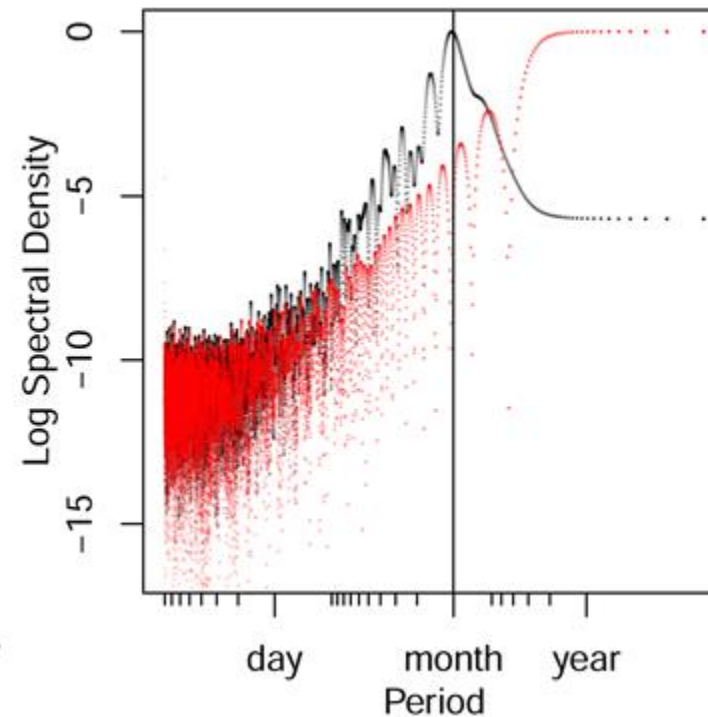
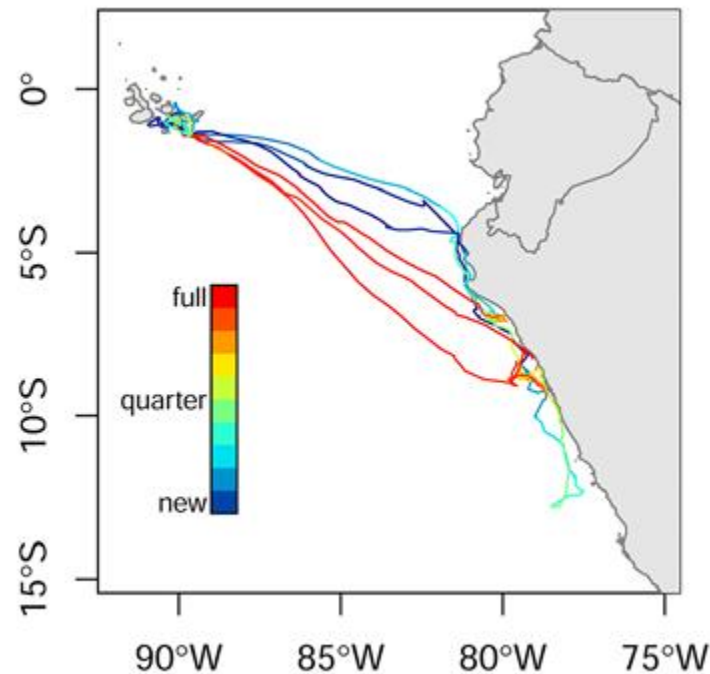


African Buffalo
(*Syncerus caffer*)



Case studies

- Waved albatross:
 - Sampling interval: 90 min



Robertson, 2011

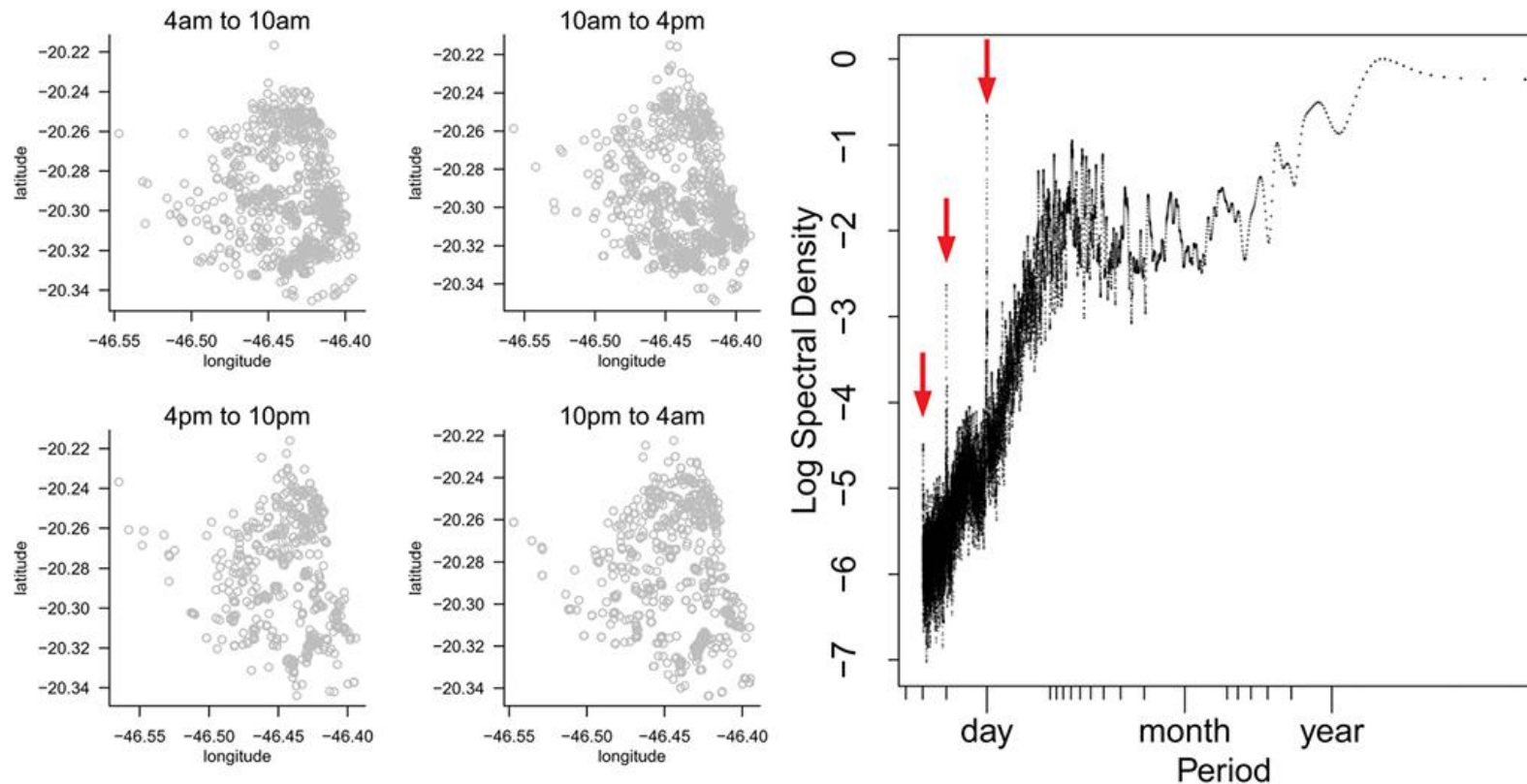
Waved albatross
(*Phoebastria irrorata*)

Case studies

- Maned wolves:
 - Sampling interval: 1-4 hours (depending on the animal)



Maned wolf
(*Chrysocyon brachyurus*)





ctmm_periodicities.R

Thank you

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